

DBSEDBD_TRIGGERS_AND_PROCEDURES

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Query 1: -CREATE DATABASE

```
1 • CREATE DATABASE Bank_ACID_DB;  
2 • USE Bank_ACID_DB;
```

✓	1	22:23:33	CREATE DATABASE Bank_ACID_DB
✓	2	22:23:37	USE Bank_ACID_DB

Query 2: - CREATE CUSTOMER TABLE

```
3 • CREATE TABLE Customer (  
4     Customer_ID INT PRIMARY KEY,  
5     Customer_Name VARCHAR(100) NOT NULL,  
6     Phone VARCHAR(15),  
7     Email VARCHAR(100),  
8     City VARCHAR(50)  
9 );
```

Output			
Action Output			
#	Time	Action	
✓ 1	19:55:43	CREATE DATABASE BankDB	
✓ 2	19:55:44	USE BankDB	
✓ 3	19:56:26	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name ...	

Query 3: - CREATE ACCOUNT TABLE

```

11 • CREATE TABLE Account (
12     Account_No INT PRIMARY KEY,
13     Customer_ID INT,
14     Account_Type VARCHAR(20),
15     Balance DECIMAL(12,2) DEFAULT 0,
16     Branch VARCHAR(50),
17
18     FOREIGN KEY (Customer_ID)
19     REFERENCES Customer(Customer_ID)
20 );

```

Output

Action Output

#	Time	Action
✓ 1	19:55:43	CREATE DATABASE BankDB
✓ 2	19:55:44	USE BankDB
✓ 3	19:56:26	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name ..
✓ 4	19:57:10	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, ..

Query 4: - CREATE TRANSACTION TABLE

```

22 • CREATE TABLE Bank_Transaction (
23     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
24     Account_No INT,
25     Transaction_Type VARCHAR(20),
26     Amount DECIMAL(12,2),
27     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
28
29     FOREIGN KEY (Account_No)
30     REFERENCES Account(Account_No)
31 );

```

Output

Action Output

#	Time	Action
✓ 1	19:55:43	CREATE DATABASE BankDB
✓ 2	19:55:44	USE BankDB
✓ 3	19:56:26	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name ...
✓ 4	19:57:10	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, ...
✓ 5	19:58:05	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_I...

Query 5: - CREATE LOAN TABLE

```
33 • CREATE TABLE Loan (  
34     Loan_ID INT PRIMARY KEY,  
35     Customer_ID INT,  
36     Loan_Type VARCHAR(30),  
37     Loan_Amount DECIMAL(12,2),  
38     Interest_Rate DECIMAL(5,2),  
39  
40     FOREIGN KEY (Customer_ID)  
41     REFERENCES Customer(Customer_ID)  
42 );
```

Output

Action Output

#	Time	Action
✓ 1	19:55:43	CREATE DATABASE BankDB
✓ 2	19:55:44	USE BankDB
✓ 3	19:56:26	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name ...
✓ 4	19:57:10	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, ...
✓ 5	19:58:05	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_I...
✓ 6	19:59:20	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan...

Query 6: - INSERT CUSTOMER DATA

```
44 • INSERT INTO Customer  
45 (Customer_ID, Customer_Name, Phone, Email, City)  
46 VALUES  
47 (101, 'Manasvi', '9876543210', 'manasvi@gmail.com', 'Hyderabad'),  
48  
49 (102, 'Akhila', '9876543211', 'akhila@gmail.com', 'Vijayawada'),  
50  
51 (103, 'Hasini', '9876543212', 'hasini@gmail.com', 'Bangalore'),  
52  
53 (104, 'Bhavana', '9876543213', 'bhavana@gmail.com', 'Chennai'),  
54  
55 (105, 'Aswitha', '9876543214', 'aswitha@gmail.com', 'Hyderabad');
```

Output

Action Output

#	Time	Action	Message
✓ 20	20:04:14	USE BankDB	0 row(s) affected
✓ 21	20:04:16	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Nam...	0 row(s) affected
✓ 22	20:04:19	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID IN...	0 row(s) affected
✓ 23	20:04:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_...	0 row(s) affected
✓ 24	20:04:27	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Lo...	0 row(s) affected
✓ 25	20:04:53	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...	5 row(s) affected

Query 7: - INSERT ACCOUNT DATA

```
57 • INSERT INTO Account
58 (Account_No, Customer_ID, Account_Type, Balance, Branch)
59 VALUES
60 (10001, 101, 'Savings', 50000, 'Hyderabad'),
61
62 (10002, 102, 'Savings', 75000, 'Vijayawada'),
63
64 (10003, 103, 'Current', 120000, 'Bangalore'),
65
66 (10004, 104, 'Savings', 45000, 'Chennai'),
67
68 (10005, 105, 'Current', 90000, 'Hyderabad');
```

Output

Action Output

#	Time	Action
✓ 21	20:04:16	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Nam...
✓ 22	20:04:19	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID IN...
✓ 23	20:04:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_...
✓ 24	20:04:27	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Lo...
✓ 25	20:04:53	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✗ 26	20:05:11	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✓ 27	20:06:22	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branc...

Query 8:- INSERT TRANSACTION DATA

```
70 • INSERT INTO Bank_Transaction
71 (Account_No, Transaction_Type, Amount)
72 VALUES
73 (10001, 'DEPOSIT', 10000),
74
75 (10002, 'DEPOSIT', 15000),
76
77 (10003, 'WITHDRAW', 20000),
78
79 (10004, 'DEPOSIT', 5000),
80
81 (10005, 'WITHDRAW', 10000);
```

Output

Action Output

#	Time	Action
✓ 22	20:04:19	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID IN...
✓ 23	20:04:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_...
✓ 24	20:04:27	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Lo...
✓ 25	20:04:53	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✗ 26	20:05:11	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✓ 27	20:06:22	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branc...
✓ 28	20:07:06	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...

Query 9: - INSERT LOAN DATA

```
83 • INSERT INTO Loan
84 (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
85 VALUES
86 (501, 101, 'Home Loan', 5000000, 7.5),
87
88 (502, 102, 'Education Loan', 1000000, 6.5),
89
90 (503, 103, 'Car Loan', 800000, 8.2),
91
92 (504, 104, 'Personal Loan', 500000, 10.5);
```

Output

Action Output

#	Time	Action
✓ 23	20:04:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_...
✓ 24	20:04:27	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Lo...
✓ 25	20:04:53	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✗ 26	20:05:11	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✓ 27	20:06:22	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branc...
✓ 28	20:07:06	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...
✓ 29	20:08:55	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_R...

Query 10: - DISPLAY TABLE DATA

```
94 • SELECT * FROM Customer;
```

95

Result Grid

Filter Rows:

Edit:

Export/Import

	Customer_ID	Customer_Name	Phone	Email	City
▶	101	Manasvi	9876543210	manasvi@gmail.com	Hyderabad
	102	Akhila	9876543211	akhila@gmail.com	Vijayawada
	103	Hasini	9876543212	hasini@gmail.com	Bangalore
	104	Bhavana	9876543213	bhavana@gmail.com	Chennai
	105	Aswitha	9876543214	aswitha@gmail.com	Hyderabad
✱	NULL	NULL	NULL	NULL	NULL

96 • `SELECT * FROM Account;`

97

Result Grid			Filter Rows:		Edit:			
	Account_No	Customer_ID	Account_Type	Balance	Branch			
▶	10001	101	Savings	50000.00	Hyderabad			
	10002	102	Savings	75000.00	Vijayawada			
	10003	103	Current	120000.00	Bangalore			
	10004	104	Savings	45000.00	Chennai			
	10005	105	Current	90000.00	Hyderabad			
✱	NULL	NULL	NULL	NULL	NULL			

98 • `SELECT * FROM Bank_Transaction;`

99

Result Grid			Filter Rows:		Edit:				Export/Import:
	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date				
▶	1	10001	DEPOSIT	10000.00	2026-08-16 20:07:06				
	2	10002	DEPOSIT	15000.00	2026-08-16 20:07:06				
	3	10003	WITHDRAW	20000.00	2026-08-16 20:07:06				
	4	10004	DEPOSIT	5000.00	2026-08-16 20:07:06				
	5	10005	WITHDRAW	10000.00	2026-08-16 20:07:06				
✱	NULL	NULL	NULL	NULL	NULL				

100 • `SELECT * FROM Loan;`

Result Grid			Filter Rows:		Edit:				Ex
	Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate				
▶	501	101	Home Loan	5000000.00	7.50				
	502	102	Education Loan	1000000.00	6.50				
	503	103	Car Loan	800000.00	8.20				
	504	104	Personal Loan	500000.00	10.50				
✱	NULL	NULL	NULL	NULL	NULL				

Query 11: -START A TRANSACTION

102	•	START TRANSACTION;
103		
Output		
Action Output		
#	Time	Action
✓ 5	22:25:26	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_...
✓ 6	22:25:29	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Lo...
✓ 7	22:25:34	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✓ 8	22:25:38	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branc...
✓ 9	22:25:41	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...
✓ 10	22:25:43	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_R...
✓ 11	22:26:44	START TRANSACTION

Query 12 : - DEPOSIT MONEY

104	•	UPDATE Account
105		SET Balance = Balance + 5000
106		WHERE Account_No = 10001;
107		
Output		
Action Output		
#	Time	Action
✓ 6	22:25:29	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Lo...
✓ 7	22:25:34	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✓ 8	22:25:38	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branc...
✓ 9	22:25:41	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...
✓ 10	22:25:43	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_R...
✓ 11	22:26:44	START TRANSACTION
✓ 12	22:27:51	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001

Query 13: -CHECK BALANCE

```
108 • SELECT *
109 FROM Account
110 WHERE Account_No = 10001;
111
```

Result Grid | Filter Rows: | Edit:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
★	NULL	NULL	NULL	NULL	NULL

Query 14: -COMMIT TRANSACTION

```
112 • COMMIT;
113
```

Output

Action Output ▼

	#	Time	Action
✓	8	22:25:38	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Bran...
✓	9	22:25:41	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALU...
✓	10	22:25:43	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_...
✓	11	22:26:44	START TRANSACTION
✓	12	22:27:51	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001
✓	13	22:28:27	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	14	22:29:02	COMMIT

Query 15: -CHECK BALANCE AFTER COMMIT

```
114 • SELECT *
115 FROM Account
116 WHERE Account_No = 10001;
117
```

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Query 16: -ROLLBACK

START TRANSACTION

```
118 • START TRANSACTION;
119
```

Output

Action Output

	#	Time	Action
✓	10	22:25:43	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_...
✓	11	22:26:44	START TRANSACTION
✓	12	22:27:51	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001
✓	13	22:28:27	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	14	22:29:02	COMMIT
✓	15	22:29:56	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	16	22:30:32	START TRANSACTION

Query 17: -WITHDRAW MONEY

```
120 • UPDATE Account
121 SET Balance = Balance - 10000
122 WHERE Account_No = 10001;
123
```

Output



Action Output

	#	Time	Action
✓	11	22:26:44	START TRANSACTION
✓	12	22:27:51	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001
✓	13	22:28:27	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	14	22:29:02	COMMIT
✓	15	22:29:56	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	16	22:30:32	START TRANSACTION
✓	17	22:31:08	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001

Query 18: -CHECK BALANCE

```
124 • SELECT *
125 FROM Account
126 WHERE Account_No = 10001;
127
```

Result Grid



Filter Rows:

Edit:



	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
★	NULL	NULL	NULL	NULL	NULL

Query 19: -ROLLBACK & CHECK BALANCE AFTER ROLLBACK

```
128 • ROLLBACK;
129 • SELECT *
130 FROM Account
131 WHERE Account_No = 10001;
```

Result Grid					
Filter Rows:					
Edit:   					
	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Query 20: -SAVEPOINT

```
133 • START TRANSACTION;
```

Output			
Action Output			
	#	Time	Action
✓	15	22:29:56	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	16	22:30:32	START TRANSACTION
✓	17	22:31:08	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001
✓	18	22:31:47	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	19	22:32:16	ROLLBACK
✓	20	22:32:30	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	21	22:33:05	START TRANSACTION

Query 21: -FIRST UPDATE

```
135 • UPDATE Account
136 SET Balance = Balance + 5000
137 WHERE Account_No = 10001;
```

Output



Action Output



	#	Time	Action
✓	16	22:30:32	START TRANSACTION
✓	17	22:31:08	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001
✓	18	22:31:47	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	19	22:32:16	ROLLBACK
✓	20	22:32:30	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	21	22:33:05	START TRANSACTION
✓	22	22:33:37	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001

Query 22: -CREATE SAVEPOINT

```
139 • SAVEPOINT Deposit1;
```

Output



Action Output



	#	Time	Action
✓	17	22:31:08	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001
✓	18	22:31:47	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	19	22:32:16	ROLLBACK
✓	20	22:32:30	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	21	22:33:05	START TRANSACTION
✓	22	22:33:37	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001
✓	23	22:34:22	SAVEPOINT Deposit1

Query 23: -SECOND UPDATE

```
141 • UPDATE Account
142 SET Balance = Balance - 3000
143 WHERE Account_No = 10002;
```

Output

 Action Output

	#	Time	Action
✓	18	22:31:47	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	19	22:32:16	ROLLBACK
✓	20	22:32:30	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	21	22:33:05	START TRANSACTION
✓	22	22:33:37	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001
✓	23	22:34:22	SAVEPOINT Deposit1
✓	24	22:35:11	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002

Query 24: -CREATE SECOND SAVEPOINT

```
145 • SAVEPOINT Withdrawal1;
```

Output

 Action Output

	#	Time	Action
✓	19	22:32:16	ROLLBACK
✓	20	22:32:30	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	21	22:33:05	START TRANSACTION
✓	22	22:33:37	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001
✓	23	22:34:22	SAVEPOINT Deposit1
✓	24	22:35:11	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002
✓	25	22:35:48	SAVEPOINT Withdrawal1

Query 25: -ROLLBACK TO SAVEPOINT

147 • `ROLLBACK TO SAVEPOINT Withdrawal1;`

148

Output



Action Output

	#	Time	Action
✓	20	22:32:30	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000
✓	21	22:33:05	START TRANSACTION
✓	22	22:33:37	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001
✓	23	22:34:22	SAVEPOINT Deposit1
✓	24	22:35:11	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002
✓	25	22:35:48	SAVEPOINT Withdrawal1
✓	26	22:36:41	ROLLBACK TO SAVEPOINT Withdrawal1

Query 26: -COMMIT

149 • `COMMIT;`

150

Output



Action Output

	#	Time	Action
✓	21	22:33:05	START TRANSACTION
✓	22	22:33:37	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001
✓	23	22:34:22	SAVEPOINT Deposit1
✓	24	22:35:11	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002
✓	25	22:35:48	SAVEPOINT Withdrawal1
✓	26	22:36:41	ROLLBACK TO SAVEPOINT Withdrawal1
✓	27	22:37:10	COMMIT

Query 27: -CHECK ACCOUNTS

151 • **SELECT ***

152 **FROM Account;**

153

Result Grid



Filter Rows:

Edit:



	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	60000.00	Hyderabad
	10002	102	Savings	72000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
•	NULL	NULL	NULL	NULL	NULL